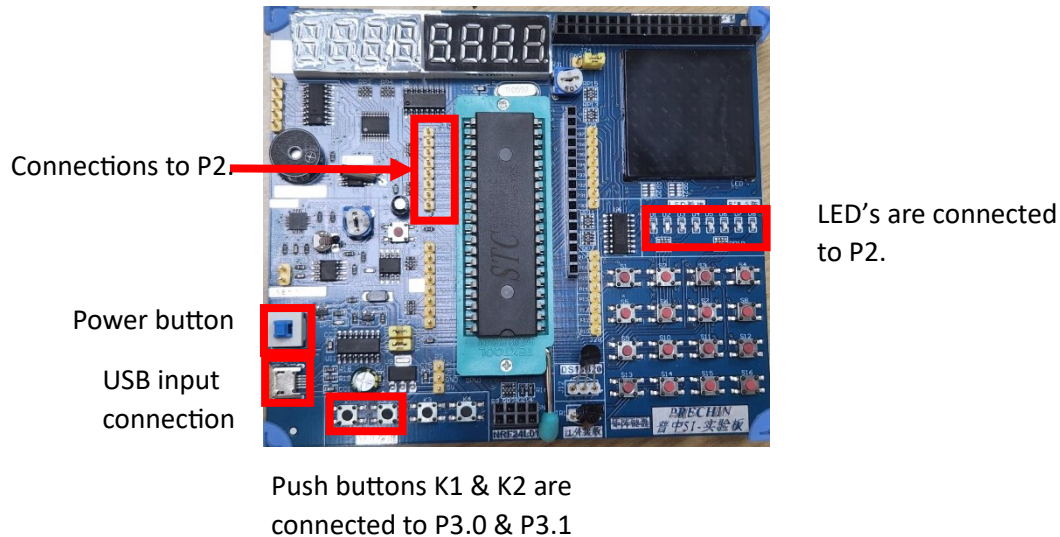


Mini Project 1

Hardware Setup:

- The 8051 Development board should have the following:
 - **8051 Microcontroller:** STC89C52
 - **Push Buttons:** Two tactile push buttons (K1 and K2)
 - **LED:** Standard LED on the board
 - (Optional) **DC Motor:** 5V DC Motor
 - **Communication & Power Supply:** USB cable



Software:

- 1) MIDE-51: the assembler that translates your assembly code to machine code. You should already have this ready.
- 2) The STC ISP tool to burn your program into the 8051 microcontroller:

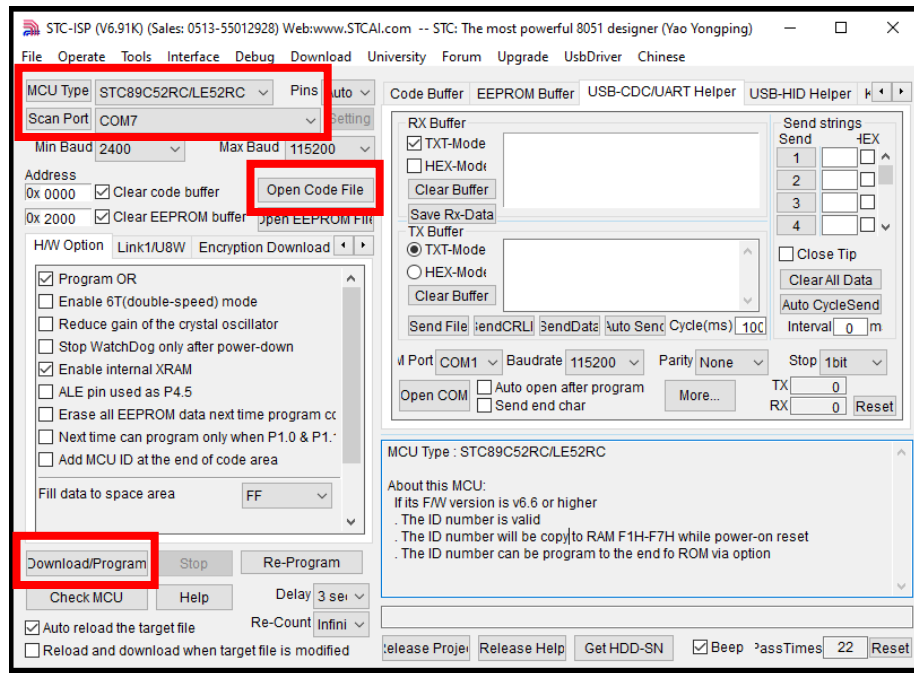
<https://www.stcmicro.com/rjxz.html>

Download the latest version of the ISP programming software. It will be a .rar extension, you can use any unzipping software to extract the file inside.

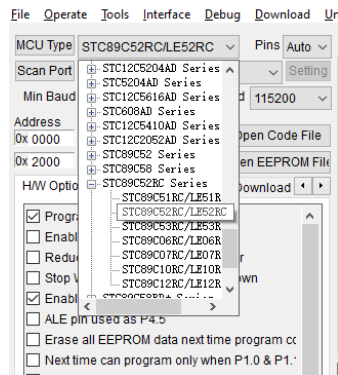
Getting Started with STC ISP to program the development board:

Connect the development board to your computer with the USB cable.

Once you open the program, you will see this window:



First, choose the correct MCU Type. Click the drop-down window and choose **STC89C52RC/LE52RC**.



Second, thing make sure you choose the correct COM port that your board is connected to (you can go to Device Manager to check which COM port is connected).

Basic program to turn on the LED:

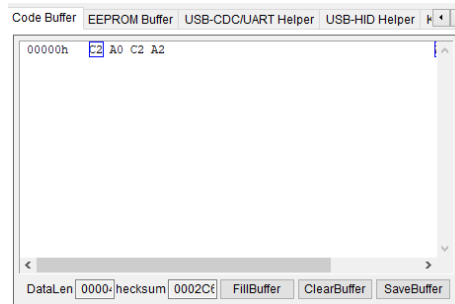
Now open **MIDE-51** and write this simple program that turns on the LED that is connected to Port 2.0:

```
CLR P2.0
END
```

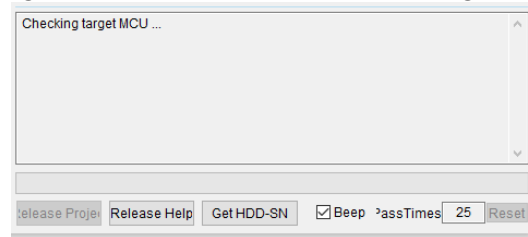
Build the code to get the machine code (which will be in a .hex file)

Go back to STC-ISP tool and click on Open Code File. Browse for the .hex file that was created by MIDE-51 (it should be in the same folder that you saved the assembly code in).

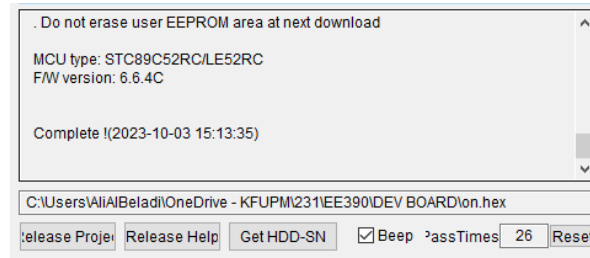
After choosing the .hex file, you should see the machine code in the Code Buffer:



Click on the Download/Program button. You should see the following on the tool:



It will not start programming the board until you power it OFF and ON again using the blue Power button on the board. Once that happens, you will see the following and your program has been successfully written onto the microcontroller.



Look at the LED's on your board. The left most LED should be on!

Now you should be ready to write the code for your mini project and upload it to the microcontroller 😊.